

SHREYA R CHITTARAGI

AI & ML Undergraduate

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ABOUT ME

AI & ML undergraduate with hands-on experience in multimodal learning, medical imaging, and deep learning systems. Skilled in building end-to-end AI pipelines including data preprocessing, model inference, explainability, and deployment using PyTorch and TensorFlow.

PROJECTS

Senthera – Multimodal Emotion Recognition System

- Designed a multimodal emotion recognition system combining text (DistilBERT), speech (Wav2Vec2), and facial expressions (ViT).
- Implemented emotion fusion logic to stabilize predictions by combining semantic and acoustic cues.
- Integrated Whisper-based real-time speech transcription with silence detection for efficient inference.
- Built an end-to-end AI system using Flask backend, supporting real-time interaction.

Parallel Multi-Model Chest X-ray Diagnosis System

- Built a modular deep learning system executing independent TB, COVID-19, and NIH multi-disease models in parallel on chest X-ray images.
- Ensured strict model isolation to prevent label interference and preserve interpretability.
- Integrated Grad-CAM explainability for clinically relevant predictions.
- Deployed a unified Streamlit-based interface for multi-model inference.

Retinal Vessel Segmentation Pipeline using MONAI

- Developed an automated medical image preprocessing and augmentation pipeline using MONAI on the DRIVE retinal dataset.
 - Applied spatial and intensity transformations to generate segmentation-ready inputs.
 - Designed the pipeline to be extensible for U-Net / Attention U-Net training with Dice/IoU evaluation.
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EXPERIENCE

AI & ML Intern — Vodafone Idea Foundation (Completed)

Completed an industry internship focused on applied AI/ML workflows, model experimentation, and real-world problem statements.

AI & ML Intern — Freelance (AQUAFROGE)

Working on AI/ML solutions under direct mentorship, focusing on data preprocessing, deep learning models, and applied AI system development.

SKILLS

- **Core ML** : Python, PyTorch, TensorFlow, scikit-learn, CNNs, Transformers
 - **Computer Vision & Medical AI** : OpenCV, YOLO, Grad-CAM, MONAI, Image Segmentation
 - **Data & Processing** : Pandas, NumPy
 - **Backend / Deployment** : Flask, REST APIs, Streamlit
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EDUCATION

B.E. in Artificial Intelligence & Machine Learning
Jawaharlal Nehru National College of Engineering
2023 – 2027

HOBBIES

Reading novels , Learning new Languages ,
Playing chess